

Prediction Record

Expected System Bottleneck

The bottleneck is expected to be the Ollama model backend. Assuming strictly sequential processing with no caching and no queuing, each candidate model's throughput ceiling is computed from its predicted single-request latency (3,600s/latency, see Per-model Predictions):

- llama3.2:1b: $3,600/7s = 514$ tickets/hr
- mistral:7b: $3,600/20s = 180$ tickets/hr
- gemma4:e4b: $3,600/45s = 80$ tickets/hr
- phi3:3.8b: $3,600/8s = 450$ tickets/hr

All models fall short of throughput requirements TR-1 average-load requirement ($\geq 1,312$ tickets/hr), and further short of TR-2 peak requirement ($\geq 3,936$ tickets/hr). We expect this ceiling to be the binding constraint under load, with latency growing without bound as arrival rate approaches it, before storage or the service layer become limiting.

Per-model Predictions

Model	Single-request latency (p95: 299 words, 1722 chars)	Overall accuracy (golden set)
llama3.2:1b	2~7s	60~62%
mistral:7b	15~20s	80~85%
gemma4:e4b	35~45s	90~95%
phi3:3.8b	3~8s	75~80%

These predictions imply two requirements failures we can check in Step 5. The llama3.2:1b predicted accuracy falls short of Accuracy Requirement AR-3: $\geq 85\%$ overall and gemma4:e4b predicted latency (35-45s) exceeds Response Time Requirement RR-1 p95 $\leq 30s$.

Hardest Prediction Categories

Debt collection and credit reporting will be the hardest based on 50% of disagreements in the golden set (from step 1). The resolution conducted in step 1 required an explicitly decision-order rule that a zero-shot model prompt won't have unless it is encoded in. Other disagreements for Money transfer and Bank account are secondary.